

Statistical methods for computer science

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Continuous Probability Distributions: Normal, Chi-square, Student-t, F

Where were we?

*Statistical science is the science of developing and applying methods for **collecting**, **analyzing** and **interpreting** data.*

- ✓ Design phase
- ✓ Description phase

Inference:

- ✓ Probability
- ✓ Introduction to random variables
- ✓ Discrete and continuous random variables
- ✓ Discrete Probability Distributions

Today Continuous Probability Distributions

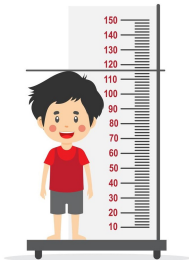
Course Outline

- Basic concepts
- Descriptive statistics (+ R Lab)
- Introduction to probability
- Introduction to random variables
- Discrete probability distributions
- **Continuous probability distributions (+ R Lab)**
- Sampling distributions
- Statistical inference: estimation (+ R Lab)
- Statistical inference: testing (+ R Lab)
- Linear model: estimation, inference, diagnostics (+ R Lab)

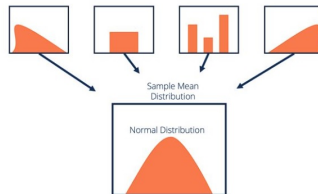
Practical Applications

Central role in a large body of statistics

Biological measurements (e.g., height, weight) approximately follow a Normal distribution



Used to approximate a large variety of distributions (theoretical justification: central limit theorem)



PDF

X is a Normal (Gaussian) RV iff its PDF is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}, \quad x \in \mathbb{R} \quad (1)$$

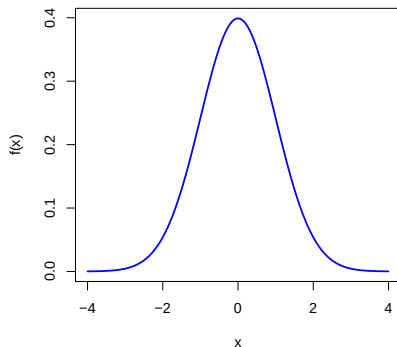
Remarks

- $X \sim N(\mu, \sigma^2)$ indicates that X is distributed as a Normal distribution with parameters μ and σ
- Domain: $\mathcal{D}_f = \mathbb{R}$, parameter space: $\mu \in \mathbb{R}, \sigma^2 > 0$
- A special case of the Normal distribution is $N(0, 1)$ whose PDF is given by:

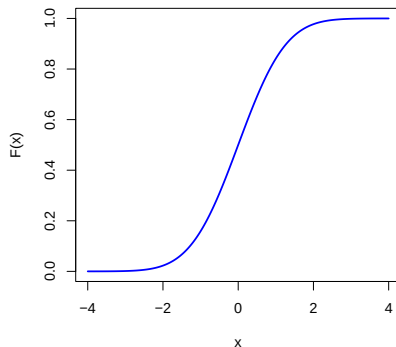
$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

PDF

Standard Normal PDF

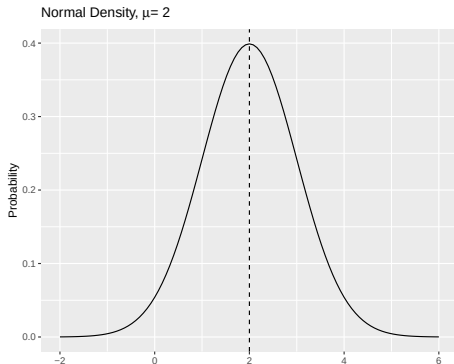


Standard Normal CDF



PDF

Main characteristics of the PDF implicit in the definition



- 1 Symmetric around μ , which is also its maximum
- 2 Two tails going asymptotically to zero
- 3 Two flex points at $x_1 = \mu - \sigma$, $x_2 = \mu + \sigma$

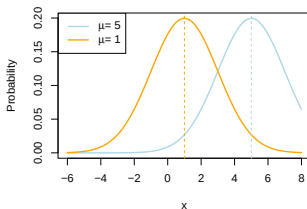
Exercise

Show 1–3 by computing first and second derivative of (4)

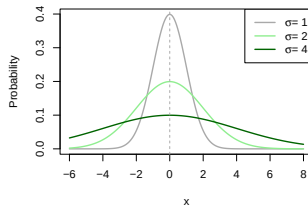
Main Expectations

Let $X \sim N(\mu, \sigma^2)$, its expected value is given by $E[X] = \mu$ and the variance is given by $V[X] = \sigma^2$.

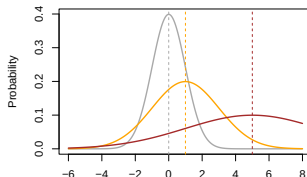
Normal distributions, different μ



Normal distributions, different σ



Normal distributions, different μ and σ



Main Properties

Linearity

$$① \quad X \sim N(\mu, \sigma^2) \Rightarrow a + bX \sim N(a + b\mu, b^2\sigma^2)$$

Normality is preserved under linear transformation

Standardization

$$② \quad X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Additivity

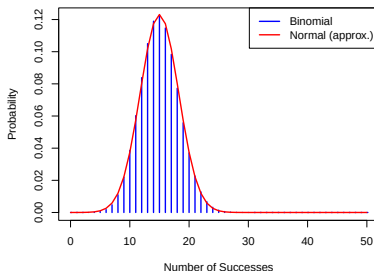
$$③ \quad X_1 \sim N(\mu_1, \sigma_1^2), X_2 \sim N(\mu_2, \sigma_2^2) \text{ and } X_1 \perp X_2 \text{ then} \\ X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Main Properties

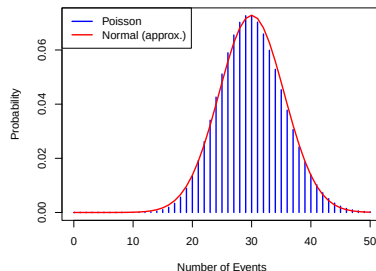
Approximations

- ④ $Bi(n, p) \approx N(\mu = np, \sigma^2 = npq)$ if $npq \geq 10$
- ⑤ $Po(\lambda) \approx N(\mu = \lambda, \sigma^2 = \lambda)$ if $\lambda \geq 10$

Binomial vs. Normal Distribution



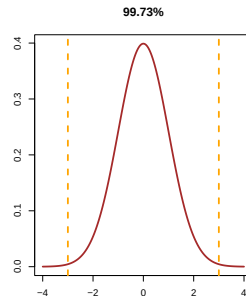
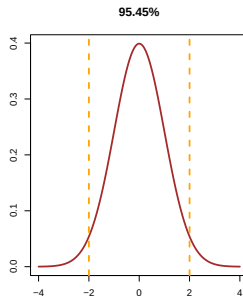
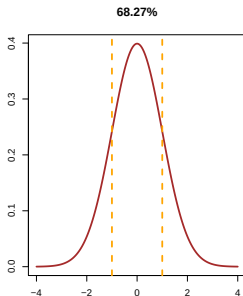
Poisson vs. Normal Distribution



Main Properties

Empirical Rule

If $X \sim N(\mu, \sigma^2)$, then $P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 68.27\%$,
 $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 95.45\%$, $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 99.73\%$



Normal Tables

- If we know that $X \sim N(\mu = 3, \sigma^2 = 4)$, how can we compute the probability $P(X \leq 2)$?
- In principle, we should calculate

$$P(X \leq 2) = \int_{-\infty}^2 \frac{1}{\sqrt{8\pi}} e^{-\frac{(x-3)^2}{8}} dx$$

- However, this integral is impossible to solve analytically (for any value of μ and σ^2) so we need to resort to statistical tables
- The tables are referred to the standard Normal distribution and usually report the values for $P(Z \leq z)$ (e.g., see [these tables](#))

Normal Tables

Example 1

Assume $Z \sim N(0, 1)$, find

① $P(Z \leq 1.23)$

② $P(Z > 1.68)$

From the standardized Normal tables we can directly find
 $P(Z \leq 1.23) = 0.89065$ and $P(Z \leq 1.68) = 0.95352$. Thus,

① $P(Z \leq 1.23) = 0.89065$

② $P(Z > 1.68) = 1 - P(Z \leq 1.68) = 1 - 0.95352 = 0.04648$

Normal Tables

Example 2

Assume $X \sim N(3, 4)$, find

- ① $P(X \leq 3)$
- ② $P(X > 3.5)$
- ③ $P(2.5 \leq X \leq 4.7)$
- ① $P(X \leq 3) = 0.5$ (by the properties of the Normal distribution)
- ② First, let's find $P(X \leq 3.5)$. By standardizing we have

$$P(X \leq 3.5) = P\left(\frac{X - 3}{2} \leq \frac{3.5 - 3}{2}\right) = P(Z \leq 0.25) = 0.59871$$

Then, $P(X > 3.5) = 1 - P(X \leq 3.5) = 1 - 0.59871 = 0.40129$

- ③ $P(2.5 \leq X \leq 4.7) = P(-0.25 \leq Z \leq 0.85) = P(Z \leq 0.85) - P(Z \leq -0.25) = 0.80234 - 0.40129 = 0.40105$

Exercise

After the release of Star Game, the company receives a substantial amount of data regarding its usage by players. These data include information on the scores achieved by the players in each game: the average score stands at 36 points with a variance of 16 (negative scores are also possible due to the penalties). Assume that the random variable describing the scores is Normally distributed with mean and variance equal to the sample mean and sample variance.

- 1 Find the probability of obtaining a score comprised between 30 and 45
- 2 How much is the third quartile of the scores?
- 3 What is the interval, symmetric around the mean, enclosing the 95% of the probability?
- 4 Plot the PDF and the CDF of the RV

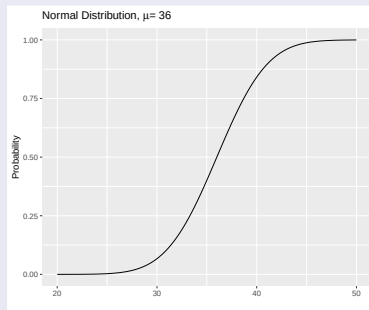
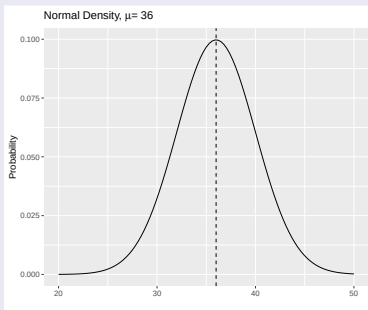
Exercise

Let X be a RV describing the scores achieved by players.

- 1 $P(30 \leq X \leq 45) = P\left(\frac{30-36}{4} \leq X \leq \frac{45-36}{4}\right) = P(-1.5 \leq Z \leq 2.25) = P(Z \leq 2.25) - P(Z \leq -1.5) = 0.98778 - 0.06681 = 0.92097$
- 2 $P(X \leq x) = 0.75 = P\left(Z \leq \frac{x-36}{4}\right) = P(Z \leq z)$ By the standard Normal tables we know that the quantile leaving a left-probability of 0.75 is approximately $z = 0.68$. Thus, we can find $x = 0.68 \cdot 4 + 36 = 38.72$
- 3 Similarly, $P(x_1 \leq X \leq x_2) = P(-z \leq Z \leq z) = P(Z \leq z) - P(Z \leq -z)$. Observe that we are now looking at a symmetric interval around 0. Thus, $P(Z \leq -z) = P(Z > z) = 1 - P(Z \leq z)$. Substituting above yields $P(-z \leq Z \leq z) = 2P(Z \leq z) - 1 = 0.95$ and $P(Z \leq z) = 0.975$. Finally, from the standard Normal tables, we know that the quantile leaving a left-probability of 0.975 is $z = 1.96$. The symmetric interval is thus given by $x_2 = 1.96 \cdot 4 + 36 = 43.84$ and $x_1 = -1.96 \cdot 4 + 36 = 28.16$

Exercise

4 Plotting



Chi-Square

- 1 PDF
- 2 Main properties
- 3 Main expectations (mean, variance)

PDF

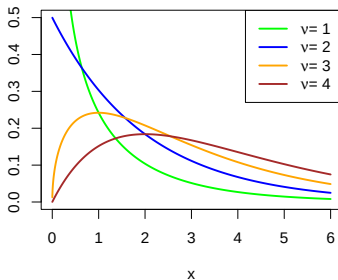
X is a Chi-square RV iff its PDF is given by

$$f_X(x) = \begin{cases} \frac{x^{\frac{\nu}{2}-1} e^{-\frac{1}{2}x}}{2^{\frac{\nu}{2}} \Gamma(\frac{\nu}{2})} & x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

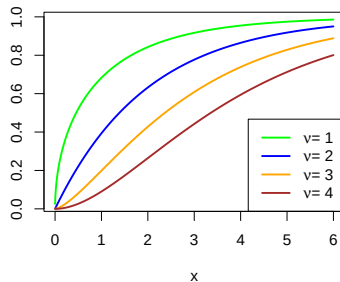
- $X \sim \chi^2(\nu)$ indicates that X is distributed as a Chi-square distribution with parameter ν
- Domain: $\mathcal{D}_f = \mathbb{R}^+ \setminus \{0\}$ (note: 0 is included only when $\nu \geq 2$)
parameter space: $\nu > 0$
- The parameter ν indicates the number of “degrees of freedom”

PDF

Chi-square PDF

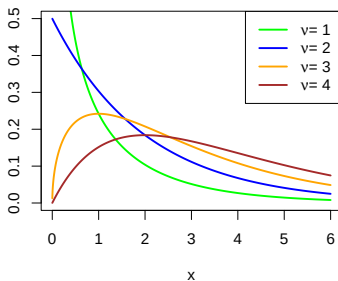


Chi-square CDF

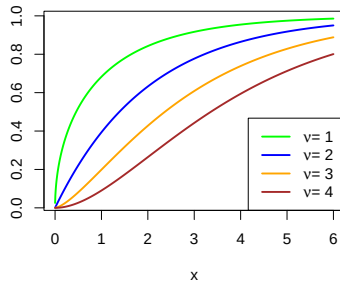


PDF

Chi-square PDF



Chi-square CDF



Main Properties

Additivity

- ① Let $X_1 \sim \chi^2(\nu_1)$, $X_2 \sim \chi^2(\nu_2)$ and $X_1 \perp X_2$
 $\Rightarrow X_1 + X_2 \sim \chi^2(\nu_1 + \nu_2)$

Link with Standard Normal I

- ② Let $Z \sim N(0, 1) \Rightarrow Z^2 \sim \chi^2(1)$

Link with Standard Normal II

- ③ Let $(Z_1, \dots, Z_n) \sim i.i.d. N(0, 1) \Rightarrow \sum_{i=1}^n Z_i^2 \sim \chi^2(n)$

Link with sample variance

- ④ Let $X_1, X_2, \dots, X_n \sim i.i.d. N(\mu, \sigma_2)$ and S_X^2 the sample variance.
Then $(n-1)S_X^2/\sigma_2 \sim \chi^2(n-1)$

Main Expectations

Let $X \sim \chi^2(\nu)$, its expected value is given by $E[X] = \nu$ and the variance is given by $V[X] = 2\nu$

Proof.

From the usual shortcut for variance computation we have

$V[Z] = E[Z^2] - E[Z]^2 \Rightarrow E[Z^2] = V[Z] + E[Z]^2 = 1$. It then follows from property 3 that

$$E[X] = E\left[\sum_{i=1}^{\nu} Z_i^2\right] = \sum_{i=1}^{\nu} E[Z_i^2] = \nu$$

Also note that $V[Z_i^2] = E[Z^4] - E[Z^2]^2 = 3 - 1 = 2$. Thus, the variance of X is given by,

$$V[X] = V\left[\sum_{i=1}^{\nu} Z_i^2\right] = \sum_{i=1}^{\nu} V[Z_i^2] = 2\nu$$

Student's t

- 1 Practical applications
- 2 PDF
- 3 Main expectations (mean, variance)
- 4 Main properties

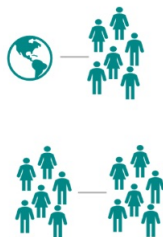
Practical Applications

Central role in a large body of statistics

Used to model heavy tailed phenomena, e.g., financial returns



Large use in frequentist inference (e.g., t-test and confidence intervals for the mean when the variance is unknown)



PDF

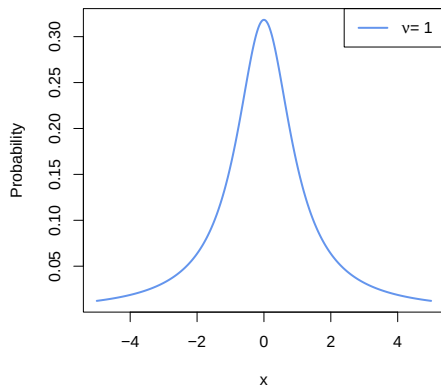
X is a Student's t RV iff its PDF is given by

$$f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}, \quad x \in \mathbb{R} \quad (3)$$

Remarks

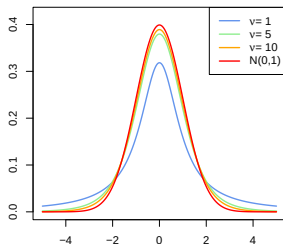
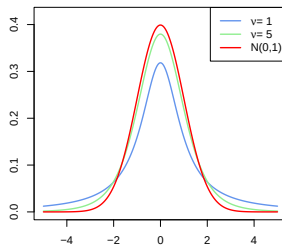
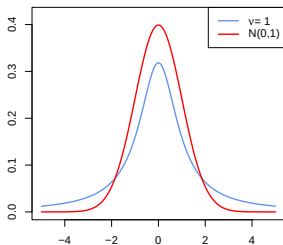
- $X \sim T(\nu)$ indicates that X is distributed as a Student's t distribution with parameter ν
- Domain: $\mathcal{D}_f = \mathbb{R}$, parameter space: $\nu > 0$
- The parameter ν indicates the number of “degrees of freedom”

PDF



- ① Bell shaped
- ② Symmetric around zero
- ③ For small values of ν the PDF has “fat tails”, when $\nu \rightarrow +\infty$ the PDF approaches a $N(0, 1)$

PDF



Main Expectations

Let $X \sim T(\nu)$, its expected value and variance are given, respectively, by

$$E[X] = \begin{cases} 0 & \text{if } \nu > 1 \\ \text{undefined} & \text{otherwise} \end{cases}$$

$$V[X] = \begin{cases} \frac{\nu}{\nu-2} & \text{if } \nu > 2 \\ \text{undefined} & \text{otherwise} \end{cases}$$

Main Properties & Special Cases

- ① Let $Z \sim N(0, 1)$ and $W \sim \chi^2(\nu)$. Then, if Z and W are independent, the variable

$$\frac{Z}{\sqrt{W/\nu}} \sim T(\nu)$$

- ② As $\nu \rightarrow +\infty$, $T(\nu) \rightarrow N(0, 1)$

- ③ $T(\nu = 1) = \text{Cauchy}(0, 1)$

The Cauchy is a RV with tails so heavy that its mean and variance do not exist

Exercise

The algorithm controlling an electronic timer for an industrial process is designed to ring when the temperature of the process reaches 120 degrees. However, due to the inherent measurement error in the temperature sensor, the timer is occasionally delayed or anticipates the actual temperature. Let's denote the random variable X as the time lags (in seconds) of the timer. Suppose these errors follow a t-distribution with 3 degrees of freedom.

- Explain why the Student's t distribution is suited to describe these errors
- Calculate the probability that the timer anticipates the actual temperature of more than 2 seconds
- Determine the value of X such that the probability of a delay of more than t seconds is 0.10.
- Find the interval, symmetric around the mean, including the 98% of the observations

Fisher-F

- 1 Practical applications
- 2 PDF
- 3 Main properties

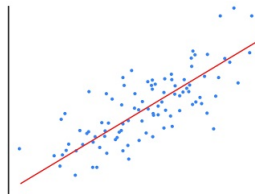
Practical Applications

It is widely used in statistical inference, but not so much for modeling natural phenomena

Test of equality between two population variances



Overall significance of a regression model



PDF

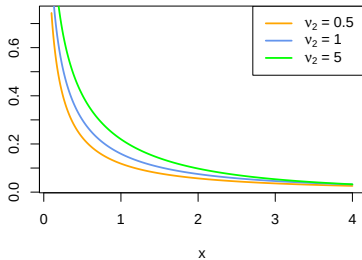
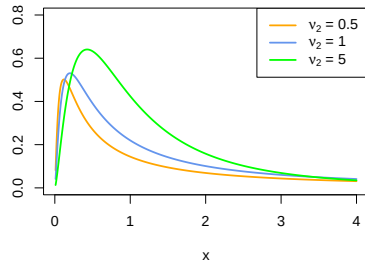
X is a Fisher-Snedecor RV iff its PDF is given by

$$f_X(x) = \frac{\Gamma\left(\frac{\nu_1 + \nu_2}{2}\right) \cdot \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} \cdot x^{\frac{\nu_1}{2} - 1}}{\Gamma\left(\frac{\nu_1}{2}\right) \cdot \Gamma\left(\frac{\nu_2}{2}\right) \cdot \left(1 + \frac{\nu_1 \cdot x}{\nu_2}\right)^{\frac{\nu_1 + \nu_2}{2}}}, \quad x > 0 \quad (4)$$

Remarks

- $X \sim F(\nu_1, \nu_2)$ indicates that X is distributed as a Fisher-Snedecor distribution with parameters ν_1, ν_2
- Domain: $\mathcal{D}_f = \mathbb{R}^+ \setminus 0$, parameter space: $\nu_1 > 0, \nu_2 > 0$
- The parameters ν_1, ν_2 both indicate the number of “degrees of freedom”

PDF

Fisher-F PDF $v_1 = 1$ Fisher-F PDF $v_1 = 5$ 

Main Properties

Link with χ^2

Let $X_1 \sim \chi^2(\nu_1)$, $X_2 \sim \chi^2(\nu_2)$ and $X_1 \perp X_2$. Then,

$$\frac{X_1/\nu_1}{X_2/\nu_2} \sim F(\nu_1, \nu_2)$$

If $X \sim F(\nu_1, \nu_2)$ then $1/X \sim F(\nu_2, \nu_1)$

If $X \sim T(\nu)$ then $X^2 \sim F(1, \nu)$